

RESEARCH SEMINAR

SimToolReal

An Object-Centric Policy for
Zero-Shot Dexterous Tool Manipulation

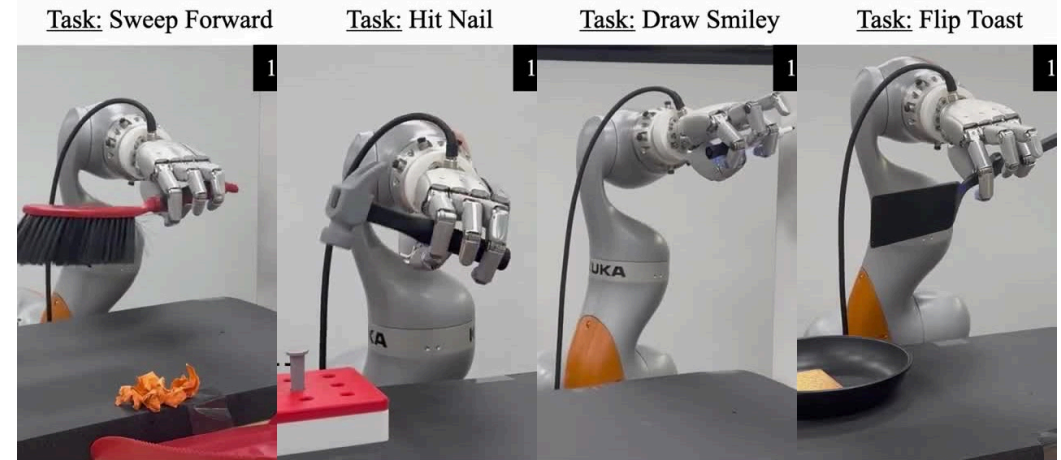
Can one reusable controller
turn tool motion into robot skill?

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How would you teach a robot to use tools?

Suppose the robot must use
a new brush, hammer or spatula.
What would you build first?

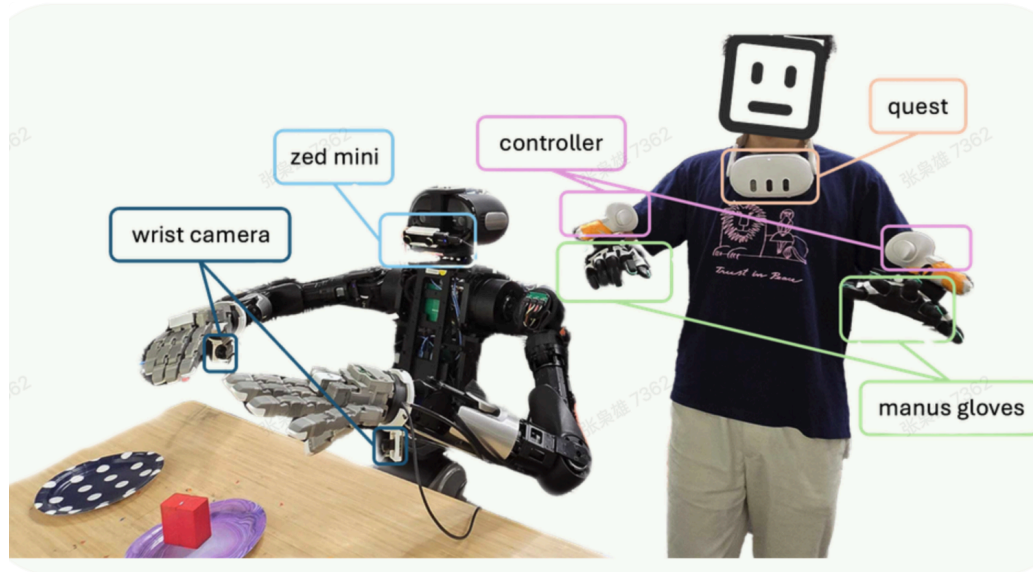
Imitate robot actions

Retarget human motion

Learn through simulation

What makes demonstration-based tool use difficult?

Robot demonstrations



Hardware-heavy capture and calibration

Human-to-robot retargeting



Human · stable wraparound grasp



Robot · infeasible grasp

A demonstrated motion does not automatically become a feasible, stable robot grasp.

Tool use as a sequence of pose-reaching problems

TASK SPECIFICATION

Human tool motion

6D pose waypoints



Reusable goal-reaching
RL policy



ROBOT EXECUTION

Arm + hand actions

Specify where the tool goes

3D position + 3D orientation

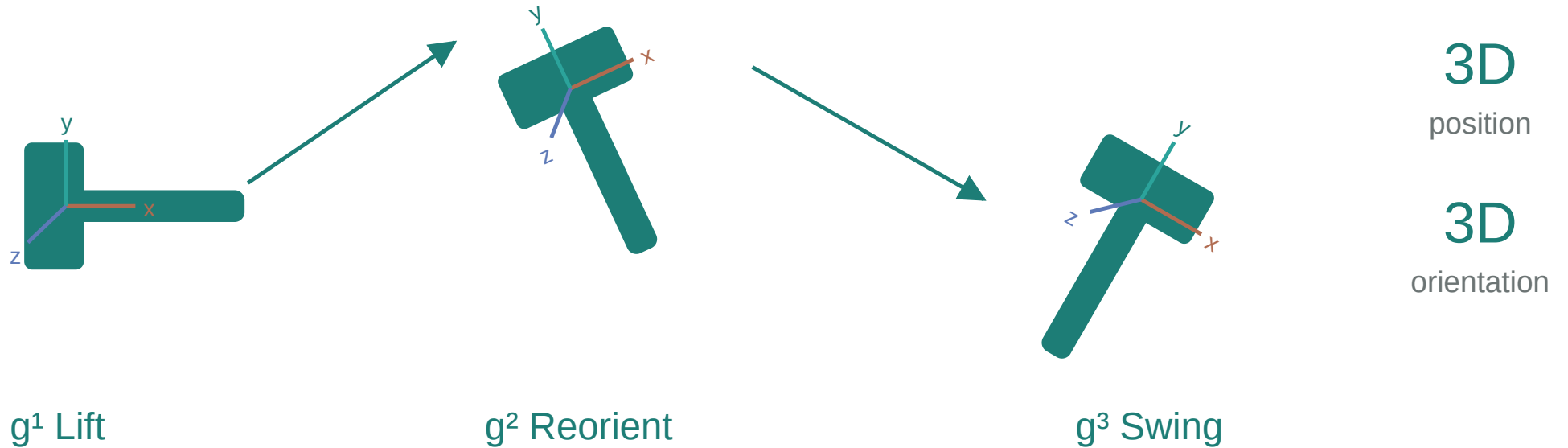
Learn how this robot moves it

Contacts and regrasping emerge from RL

Object goals specify the motion. RL discovers robot-specific grasps and in-hand actions.

A 6D pose specifies position and orientation

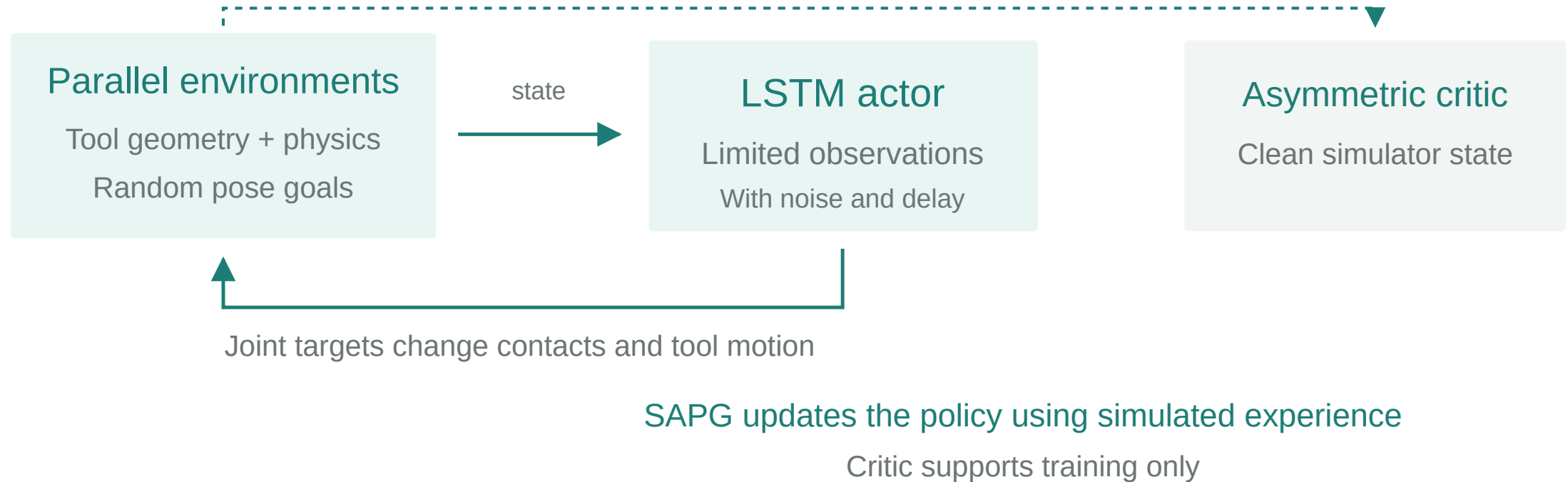
A task becomes an ordered set of rigid-object configurations



$$g^k = (R^k, p^k) \in SE(3), \quad \mathcal{G} = \{g^1, \dots, g^K\}$$

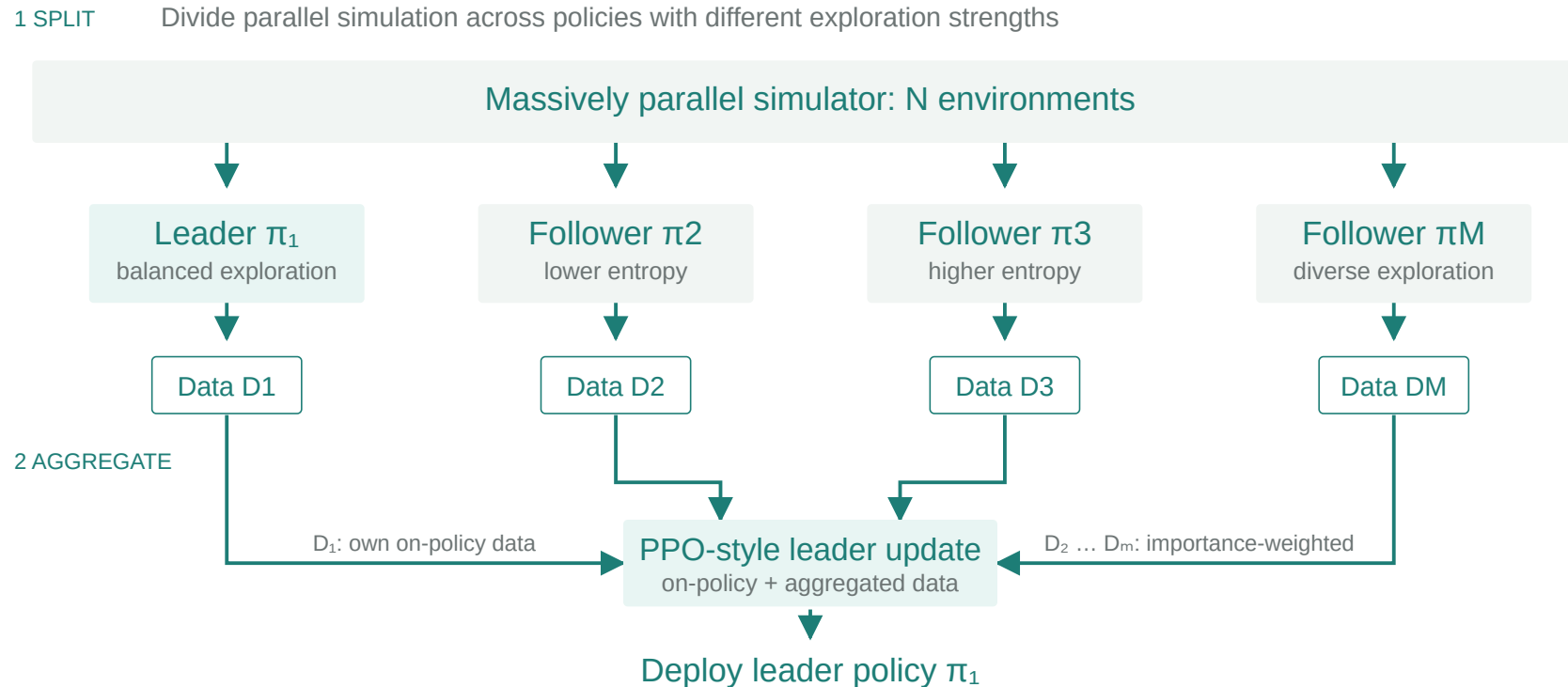
Training pipeline: learn pose reaching through interaction

SIMULATION ONLY



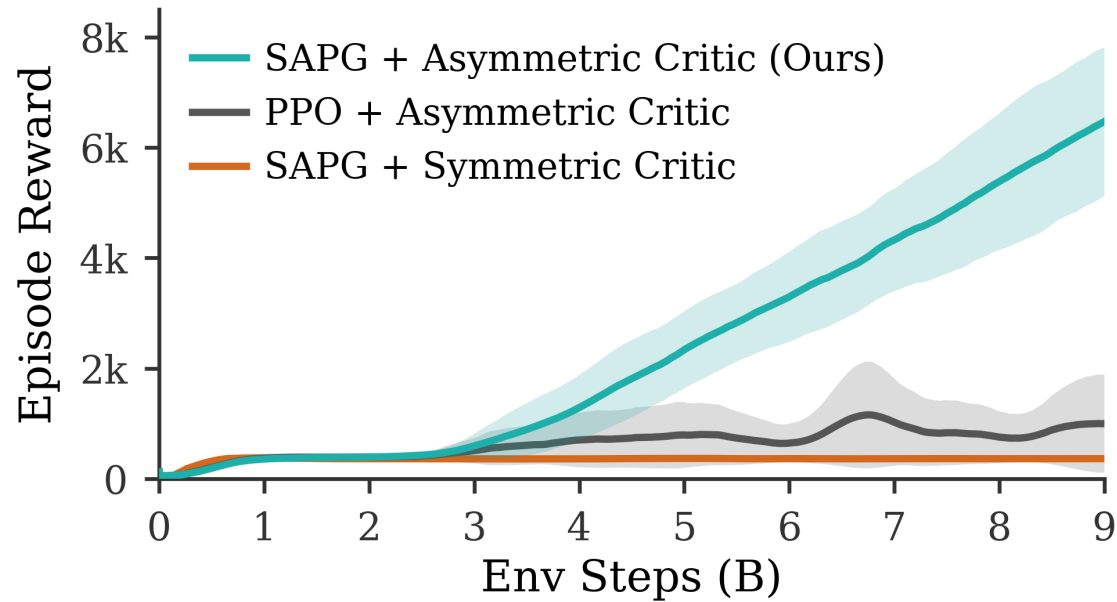
SAPG splits exploration and aggregates experience

One PPO policy can collect similar experience across thousands of parallel environments.



SAPG retains PPO-style actor–critic updates while changing how experience is explored and reused.

SAPG learns more effectively than PPO in this setting



Paper Fig. 8 · Training reward, mean over five seeds

Controlled comparison

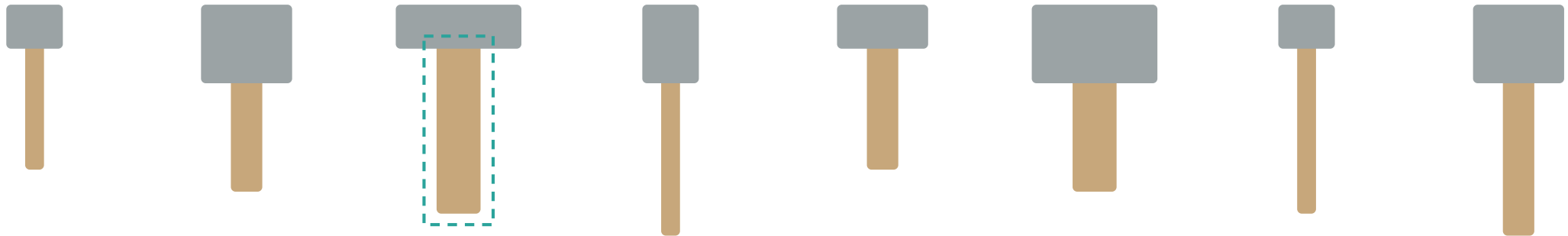
- Both runs retain the asymmetric critic.
- Replacing SAPG with PPO changes the exploration and data aggregation scheme.

Observed result

- SAPG continues improving across 9B environment steps.
- PPO reaches a much lower plateau under the matched setup.

Procedural tools cover shape and mass variation

One structural prior, many physical instances



Geometry

Shape + size

Mass distribution

Handle / head densities

Shared policy inputs

Pose + grasp-region box

The commonality is a geometric interface: tool pose and a graspable region.

RL training signal: how each reward term is constructed

$$r = r_{\text{smooth}} + r_{\text{grasp}} + \mathbb{I}_{\text{grasped}} r_{\text{goal}}$$

1 · Smooth motion

$$-\lambda_a \|\dot{q}_a\|_1 - \lambda_h \|\dot{q}_h\|_1$$

Penalize current arm and hand joint speeds.

2 · Acquire the grasp

$$r_{\text{approach}} + (1 - \mathbb{I}_{\text{grasped}}) r_{\text{lift}}$$

Approach: best-so-far fingertip distance improvement.

Lift: height progress plus a one-time bonus.

3 · Reach each goal pose

$$\Delta d^* + B_{\text{succ}} \mathbb{I}[d(o_t, g) < \epsilon]$$

Progress: improve the four-keypoint pose error.

Success: cross ϵ , receive a bonus, then sample the next goal.

These are environment reward terms. SAPG still optimizes an actor–critic objective from the resulting returns.

LSTM memory carries interaction history across steps

THE SAME RECURRENT NETWORK AT EVERY CONTROL STEP



Cell state c_t : internal memory

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t$$

Forget gate f_t retains old information. Input gate i_t writes candidate memory.

Hidden state h_t : current representation

$$h_t = o_t \odot \tanh(c_t)$$

Output gate o_t exposes information to the action MLP and the next time step.

The action head controls the arm and hand differently

$$a_t = [a_t^{\text{arm}}, a_t^{\text{hand}}] \in [-1, 1]^{29}$$

7 arm joints: incremental targets

$$\hat{q}_t^{\text{arm}} = q_{t-1}^{\text{target, arm}} + 0.025a_t^{\text{arm}}$$

Each action adjusts the previous target.

22 hand joints: absolute targets

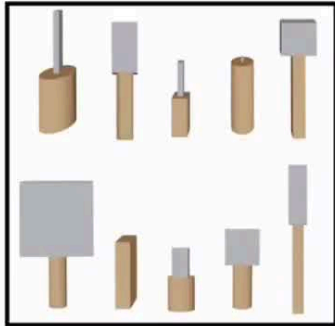
$$\hat{q}_t^{\text{hand}} = q_{\min} + \frac{a_t^{\text{hand}} + 1}{2} \odot (q_{\max} - q_{\min})$$

Each action maps into the joint's allowed range.

Joint limits and EMA smoothing ($\alpha = 0.1$) produce the final targets at 60 Hz.

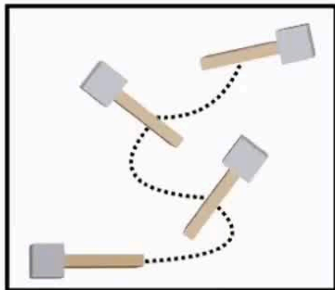
Simulation: reaching diverse tool poses

Procedurally-Generated
Primitive Objects

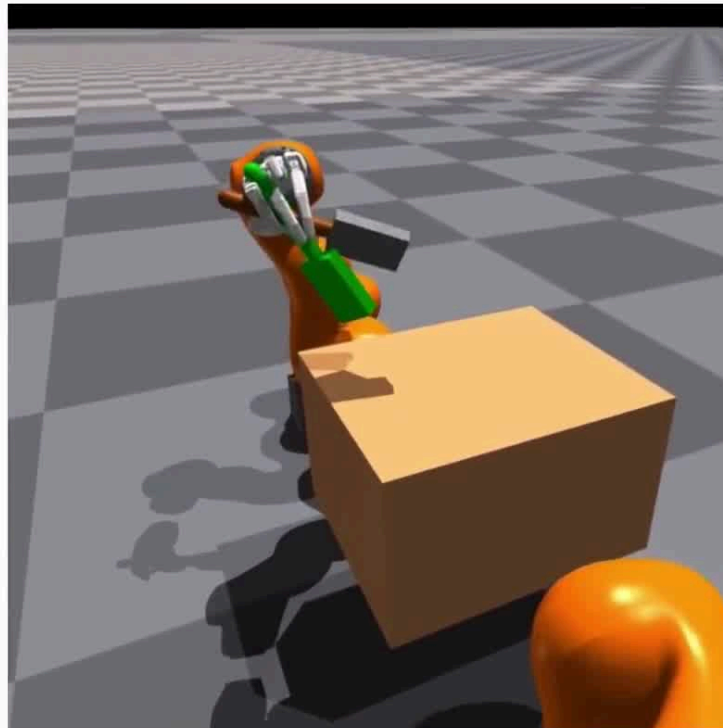


+

Random Goal Poses



Train a Single RL policy across All Objects with
Goal-Pose Reaching Rewards



WHAT TO WATCH

The **green mesh** indicates the current target pose.

The policy learns to acquire and reorient procedurally generated tools.

There is no human task trajectory in this training loop.

Simulation exposes the state. Reality requires estimation.

Inside the simulator

- Exact tool pose and geometry are available.
- Random goal poses define the task.
- Noise and delays train the actor on imperfect inputs.

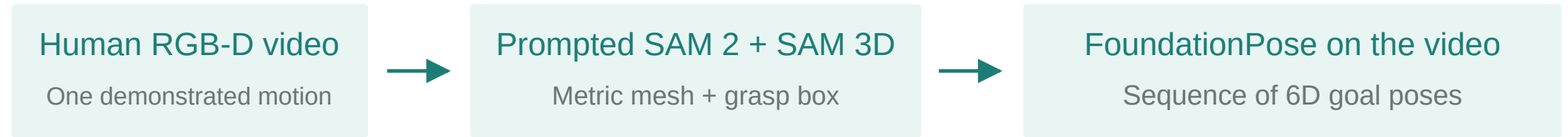
On the real robot

- The tool pose is hidden behind pixels and occlusion.
- A human demonstration supplies a meaningful trajectory.
- Perception must recover the same policy interface.

Keep the motor policy fixed. Replace simulator-derived inputs with measured and estimated inputs.

Real deployment supplies the same policy interface

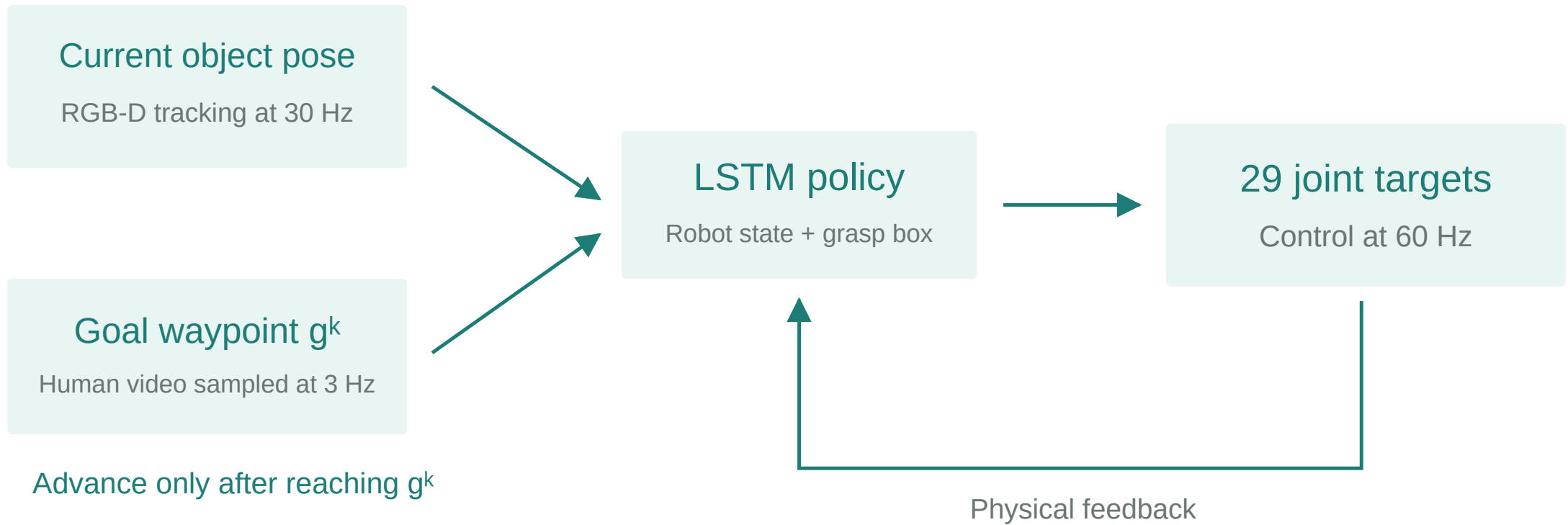
SETUP FOR A NEW TOOL / DEMONSTRATION



ONLINE EXECUTION

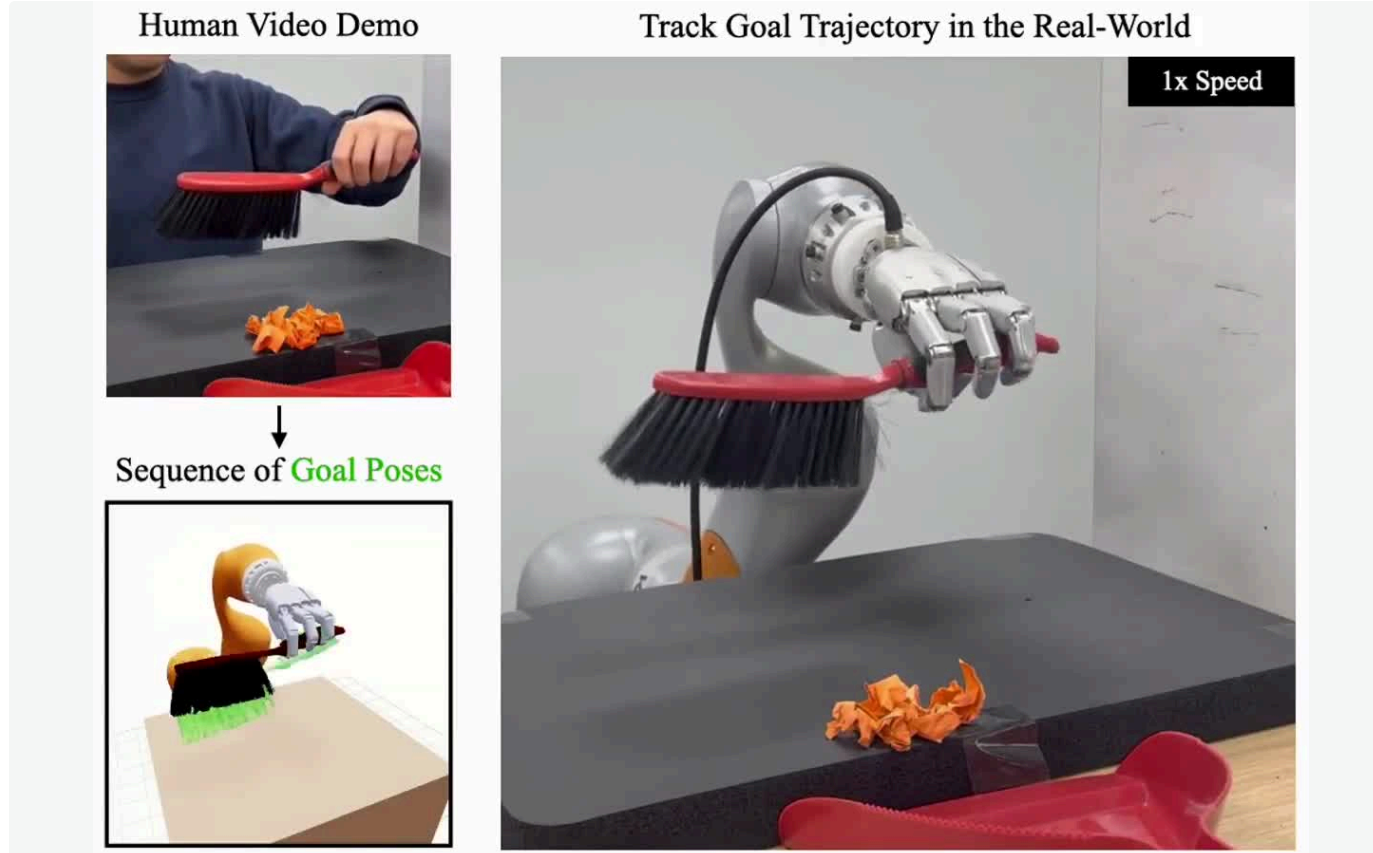


The robot advances goals only after reaching them



3 Hz waypoints · **event-driven** goal switch · 60 Hz control

The closed loop follows unseen tool trajectories



WHAT TO WATCH

The human demonstration supplies the **tool-motion goals**.

The fixed policy grasps, reorients and moves the real brush.

The goal interface specifies tool motion rather than robot finger trajectories.

Brush and hammer: reorientation before interaction

Brush

Brush (Zero-Shot)

Task: Sweep Forward



Task: Sweep Right



Reorient the handle, then sweep along the surface.

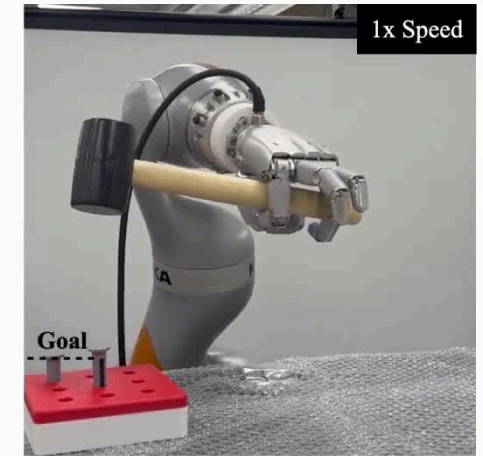
Hammer

Hammer (Zero-Shot)

Task: Hit Nail



Task: Hit Nail



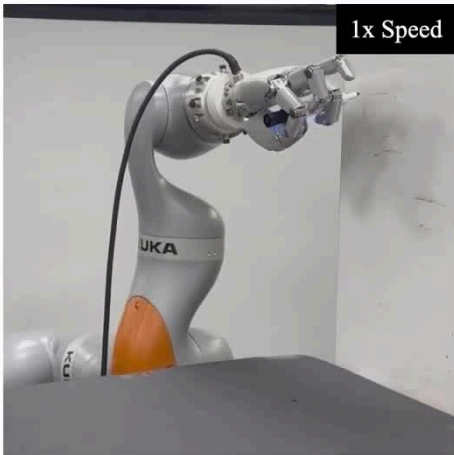
Change the grasp-tool orientation before the swing.

Marker and eraser: tool motion along a surface

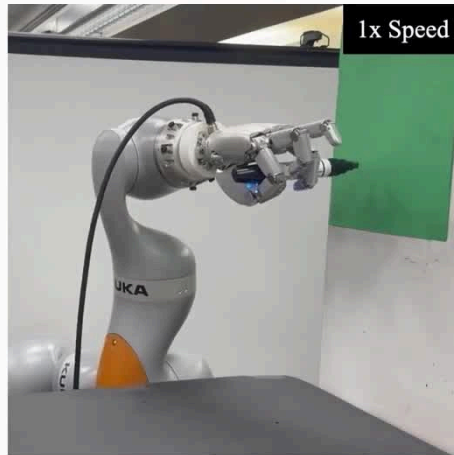
Marker

Marker (Zero-Shot)

Task: Draw Smiley



Task: Draw Smiley

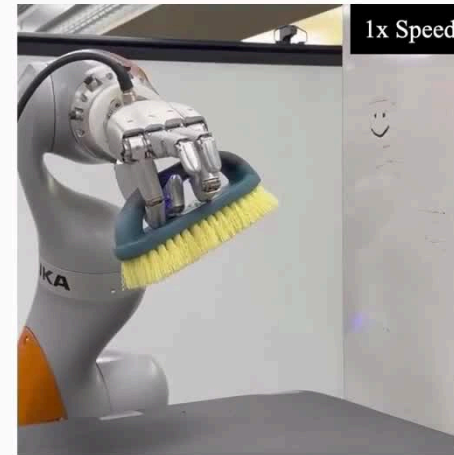


Acquire a thin tool and move its tip across the board.

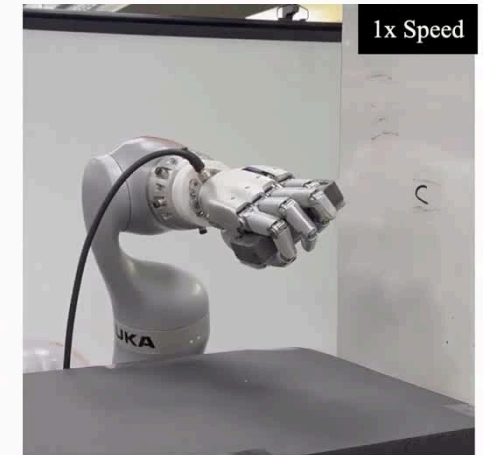
Eraser

Eraser (Zero-Shot)

Task: Wipe Whiteboard



Task: Wipe Whiteboard



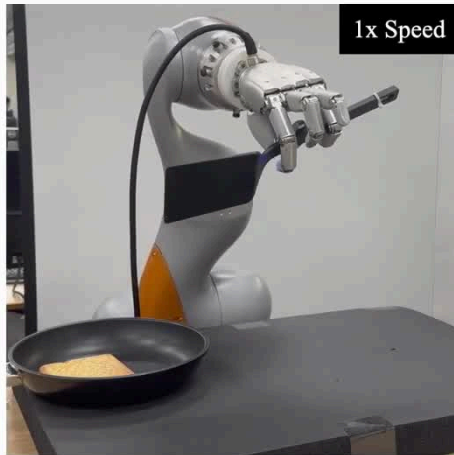
Maintain a grasp while following a wiping path.

Spatula and screwdriver: larger orientation changes

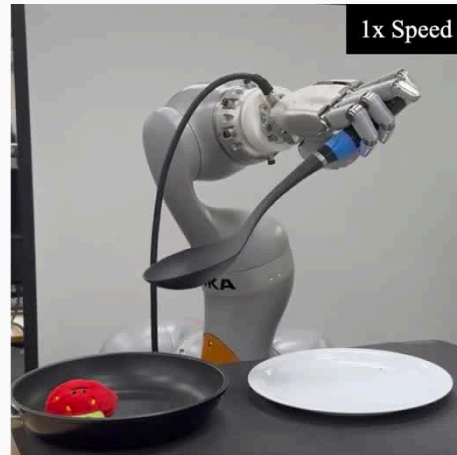
Spatula

Spatula (Zero-Shot)

Task: Flip Toast



Task: Serve Strawberry

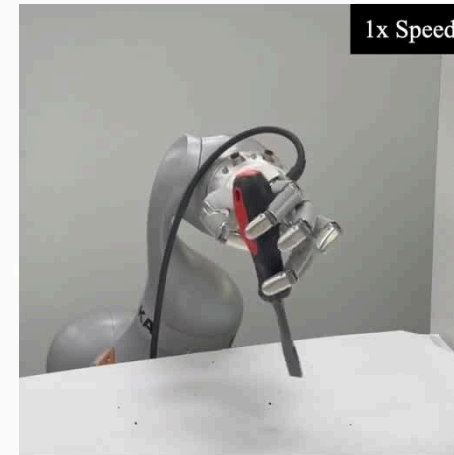


Serving and flipping demand changing tool orientation.

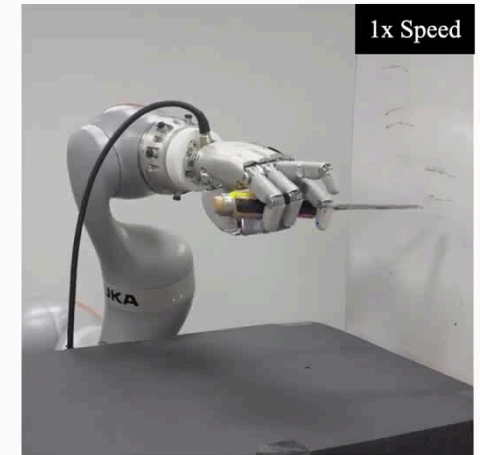
Screwdriver

Screwdriver (Zero-Shot)

Task: Spin Vertically



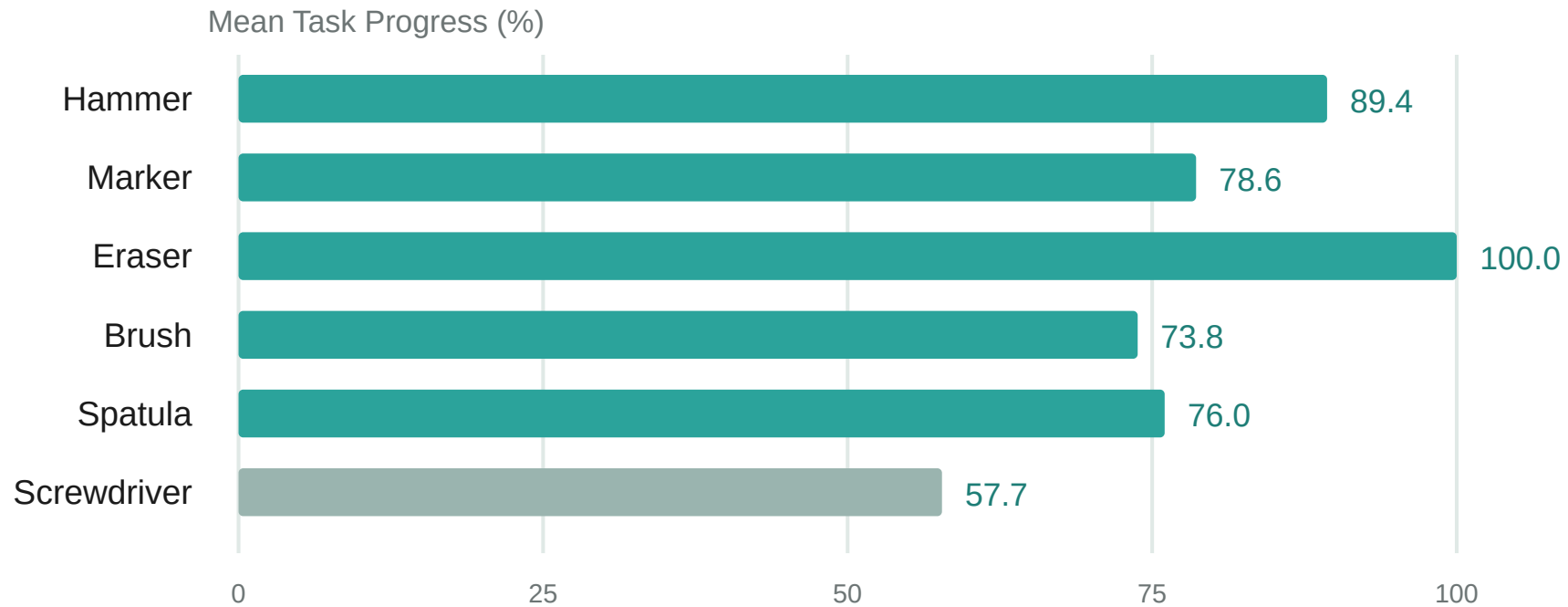
Task: Spin Horizontally



Reorient and spin the tool in free space.

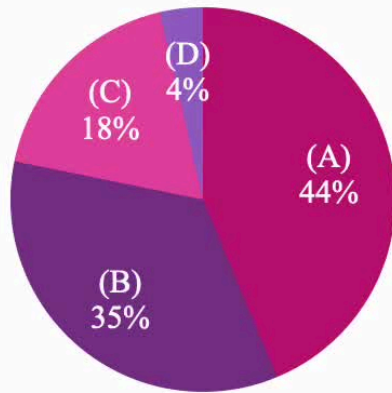
Real-world transfer across six tool categories

120 rollouts · Task Progress = pose waypoints reached



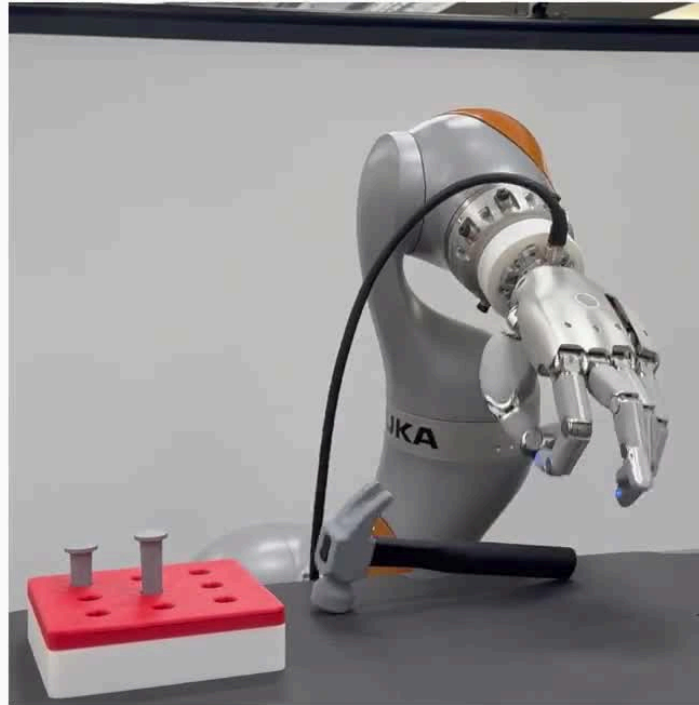
79.3% mean trajectory progress across six categories.

Recovery illustrates feedback, with clear limits



- (A) Pose Tracking Failure
- (B) Object Drop
- (C) Incomplete Rotation
- (D) Failed Grasp

Recovery from Failed Grasp



WHAT TO WATCH

The policy can attempt to **regrasp** after an error.

Recovery depends on the tool staying **reachable and trackable**.

Rigid pose goals leave force objectives, clutter and task replanning unresolved.

Structured simulation as a route to reusable robot skills

PRESENTER REFLECTION

01 · PRIMARY TAKEAWAY

Beyond scaling end-to-end action models, **structured simulation** may offer another path toward skill acquisition in the wild.

RL learns reusable motor competence. Perception, demonstrations or a higher-level model specify what the tool should do.

02 Compact interfaces beyond tool use

Search for compact task interfaces—adding contact, force or articulation when pose is insufficient.

03 Lightweight VLMs for semantic setup

Use a VLM for tool selection and subgoals; preserve geometric tracking for metric feedback.

BACKUP

References, videos and source material

Main paper

Kedia, Lum, Bohg and Liu. *SimToolReal: An Object-Centric Policy for Zero-Shot Dexterous Tool Manipulation*.
arxiv.org/abs/2602.16863v2 · [RSS 2026 proceedings](#)

Project and code

simtoolreal.github.io · github.com/tylerlum/simtoolreal

Closely related work

Chen et al. *Object-Centric Dexterous Manipulation from Human Motion Data*. CoRL 2024.

Lum et al. *Crossing the Human-Robot Embodiment Gap with Sim-to-Real RL Using One Human Demonstration*. 2025.

Singla et al. *Split and Aggregate Policy Gradients*. ICML 2024.

Petrenko et al. *DexPBT*. 2023.

Design reference

[Mizoreww/steerable-vla-talk](https://mizoreww.github.io/steerable-vla-talk)